

Himanshu Menghani

Full Stack Developer & AI Engineer

📍 Mumbai, India — 📞 +91 77699 49282 — ✉ himanshumenghani524@gmail.com —  [linkedin.com/in/himanshumenghani524](https://www.linkedin.com/in/himanshumenghani524) —  github.com/524himanshu —  <https://himanshu-menghani.vercel.app>

Summary

Full-stack Developer and AI Engineer with hands-on experience building and deploying production-ready AI and web applications using **FastAPI, Django, Next.js, React, Python, PostgreSQL, Redis, and LLMs**. Built **4+ production-oriented AI and software engineering projects** involving LLM integrations, role-based systems, asynchronous workflows, and scalable backend architectures. Developed **DrishtiAI**, shortlisted for **AI for Bharat 2026**, and **RecruitIQ**, an AI-powered candidate ranking system evaluated on **100,000 candidate profiles**. Experienced in **Agentic AI, RAG pipelines, REST API development, LLM integrations (Gemini, Claude, Llama 3.3), asynchronous workflows, Celery, Redis, and scalable backend architectures**. Focused on designing reliable, maintainable, and production-ready systems from development to deployment.

Technical Skills

Languages: JavaScript, Python, SQL

AI/ML: LLMs (Gemini, Claude, Llama 3.3), Prompt Engineering, RAG, Agentic AI, Scikit-learn, Pandas, NumPy

Backend: FastAPI, Django, Django REST Framework, Flask, Node.js, Express.js, REST APIs, JWT Authentication, Celery, Redis

Frontend: React.js, Next.js, Tailwind CSS

Databases: MongoDB, PostgreSQL

Tools & Platforms: Redis, Docker, Git, Postman, Vercel, Render, Cloudflare Workers

Concepts: RBAC, Async Processing, API Design

Projects

DrishtiAI — Real-Time Adverse Event Detection | *FastAPI, Next.js, PostgreSQL, Redis, scispaCy, Docker*

 [GitHub](#)

- Built a **real-time pharmacovigilance platform** ingesting Twitter and Reddit data, detecting adverse drug events with **94% classification accuracy** using **scispaCy NLP pipelines** and **SHAP explainability**.
- Architected a **full NLP pipeline** covering entity extraction, signal classification, and confidence scoring across live social media streams with sub-second processing.
- Designed **PostgreSQL schema** for structured signal storage with Redis-backed caching, **reducing query response latencies by 75% (from 400ms to 90ms)** under heavy loads.
- Implemented real-time **Presidio-based PII redaction** to mask patient credentials before database insertion, introducing **15ms processing overhead** to ensure HIPAA compliance.
- Deployed end-to-end on **Railway** using Docker Compose, with a **Next.js dashboard** for real-time signal monitoring and filtering.
- **Shortlisted for the Prototype Phase of AI for Bharat 2026**, a national AI innovation hackathon focused on real-world problem statements.

CodeMate — AI-Powered Interactive Learning OS | *Next.js, FastAPI, WebSockets, Python AST, Gemini API*  [GitHub](#)  [Live Demo](#)

- Designed a decoupled Next.js & FastAPI architecture featuring a multilingual compilation sandbox executing Python, JS, C++, and Java in isolated subprocesses under strict 2-second timeouts.
- Engineered a low-level Python runtime execution tracer using custom **sys.settrace** hooks, capturing frame states deterministically to replay visual call stacks on an interactive whiteboard canvas.
- Implemented collaborative multiplayer mock interview rooms utilizing FastAPI WebSockets to sync code buffers and terminal outputs instantly with **50ms network synchronization latency**.
- Built a secure compilation scanner intercepting and blocking OS exploit vectors (e.g. **system()**, **ProcessBuilder**, **import os**) across 4 languages to secure host environments.

RecruitIQ — Intelligent Candidate Ranking | *Python, Streamlit, Sentence Transformers, Pandas*

 [GitHub](#)  [Live Demo](#)

- Built an AI-powered candidate ranking system for the Redro India Runs Hackathon, evaluating **100,000 candidate profiles** using a **5-factor scoring model** covering semantic relevance, career trajectory, skills alignment, behavioral signals, and education quality.
- Implemented semantic matching with **all-MiniLM-L6-v2** embeddings and cosine similarity, **reducing recruiter screening cycles by 90%** via cached vector search completing in **60 seconds**.
- Developed profile quality checks to detect inconsistent experience histories, skill inflation, and keyword stuffing, improving ranking reliability and reducing manipulation of candidate scores.
- Submitted under **Track 1: Data & AI Challenge**; generated validator-compliant ranked outputs and successfully passed the official evaluation pipeline.

- Built an **asynchronous payout processing system** using **Celery and Redis**, offloading heavy background tasks and **reducing backend request response times by 65%**.
- Implemented **background job queues with custom retry logic and dead-letter queues**, achieving **99.98% transaction delivery reliability** on payout tasks.
- Designed **PostgreSQL transactional database workflows** to maintain strict data consistency and prevent duplicate or failed payouts.
- Integrated **HubSpot API** to sync payout and user data with external CRM workflows, handling API responses and failure scenarios.
- Developed **REST APIs using Django REST Framework** for payout initiation, status tracking, and admin operations.
- Built **RBAC-based dashboards** enabling real-time payout monitoring for admins and end users.

Professional Experience

OctaNet Services Pvt Ltd. | *Web Development Intern*

Mumbai, India | **Sept 2023 – Oct 2023**

- Developed and delivered **3+ responsive web applications**, including a task management platform featuring prioritization, deadline tracking, and status workflows, meeting all project milestones within the internship duration.
- Optimized layout responsiveness and cross-device compatibility, **reducing cumulative layout shifts (CLS) by 40%** on mobile browsers.
- Resolved rendering bottlenecks and DOM leaks, **improving frontend page-load speeds by 20%** on low-end devices.

Education

B.Sc, Artificial Intelligence and Data Science
BITS Pilani Digital

2026 – Present

Bachelor of Engineering, Information Technology
Mumbai University

2021 – 2024

Certifications & Achievements

- **AI for Bharat 2026** — Shortlisted, Prototype Phase (DrishtiAI)
- **Develop GenAI Apps with Gemini and Streamlit** — Google Cloud
- **Prompt Design in Vertex AI** — Google
- **INVICTUS'24 Hackathon** — QuestIT Cell × ISTE VESIT
- **Introduction to Generative AI** — Google
- **Cloud & Full-Stack Credentials** — Professional certifications across Serverless architectures, API design, and cloud deployments ([Credentials on LinkedIn](#))